

DECK 12 · WEEK 6 · MORNING 1

Equity II

Analysis & Valuation

From business model to intrinsic value: the five valuation tools every analyst owns

CFA Level I · Vol 5 Learning Modules 5–8

Roadmap — Equity II — Analysis & Valuation

4 sections · ~2 hours · Built bottom-up from CFA readings

01

Company analysis — past & present

Research-report anatomy · business model · revenue drivers · pricing power · operating & financial leverage

02

Industry & competitive analysis

Classification (GICS / ICB / TRBC) · Porter Five Forces · how to rank them · PESTLE · three generic strategies

03

Company forecasting

Four forecast approaches · top-down vs bottom-up revenue · scenario analysis (bear/base/bull) · iPad-vs-PC worked case

04

Valuation toolkit

DDM, Gordon & two-stage · FCFE · P/E, P/S, P/B · EV/EBITDA · asset-based — Siemens, IBM, Nestlé, Cameco

Learning objectives

By the end of this deck you should be able to:

1. Describe the eight elements of an initial equity research report, and evaluate a company's business model, revenue drivers, and operating/financial leverage.
2. Analyze an industry's structure using Porter's Five Forces and identify the dominant forces to monitor, and complement the analysis with a PESTLE scan.
3. Apply the four forecast approaches to build a top-down or bottom-up revenue forecast and quantify key risks with bear / base / bull scenarios.
4. Calculate intrinsic value using the Gordon growth model, a two-stage DDM, and an FCFE model, and identify the company profile each is suited to.
5. Compute and interpret P/E, P/S, P/B, EV/EBITDA and asset-based valuations — keeping numerator and denominator on the same claim — and understand why a low multiple demands a "why" before a "buy".

SECTION 01

Company analysis past & present

An initial equity research report sits on three analytical pillars — company, industry, and forecast — all feeding valuation and a recommendation

Initial reports cover all eight elements below. Subsequent reports are shorter and revisit only what changed — usually the financial model and the valuation.

ANALYTICAL FRAMEWORK

3. FORECASTING

Forecast objects & approaches · revenue · profitability & working capital · capital investments & structure · key risks

2. INDUSTRY & COMPETITIVE

Define industry · size, growth, profitability, share · structure (Porter) & external influences (PESTLE) · position

1. COMPANY — PAST & PRESENT

Business model · revenue & revenue drivers · operating profitability & working capital · capital investments & structure

INITIAL-REPORT ELEMENTS

1. Front matter — identifiers, rating, disclosures
2. Recommendation + target price, and why
3. Company description & business model
4. Industry overview & competitive positioning
5. Financial analysis & model
6. Valuation — relative and present-value
7. ESG — incl. ownership, management, pay
8. Material upside / downside risks

Source: CFA Institute, Curriculum Vol 5, Learning Module 5, §2 "Company Research Reports", Exhibits 1–2, pp. 202–203.

A business model is five questions — what, to whom, through which channel, at what price, and bought from whom

Verify answers against filings, earnings calls, investor events, site visits and third-party research — never management alone.

1 **WHAT IS THE FIRM SELLING?**

Products and services; recurring vs one-off; single-segment (Warehouse Club Inc.) or a conglomerate. Most firms fit a conventional model — analyse the differences.

2 **TO WHOM DOES IT SELL?**

Customers and key customer groups: one dominant buyer, a fragmented consumer base, or a captive government buyer. Concentration is the number to find.

3 **HOW DOES IT REACH THEM?**

Sales channels: customer acquisition and delivery. Own stores, wholesale, direct-to-consumer, marketplace, distributor — each carries a different cost and margin.

4 **HOW IS IT PRICED AND PAID?**

Price levels and structure, payment terms, subscriptions vs transactions. Sets the revenue line and the working-capital cycle at the same time.

5 **WHAT DOES IT BUY AND RELY ON?**

Resources, suppliers and partners. Are inputs specialised or hard to replace? This is where supplier power and single points of failure hide.

¹ Competitive strategy is not one of the five elements — it belongs to industry analysis (LM 6, §6). Describe first, judge later.

Source: CFA Institute, Curriculum Vol 5, Learning Module 5, §3 "Determining the Business Model", pp. 205–206.

Revenue analysis decomposes growth into volumes, prices and mix – with pricing power the decisive lever

Pricing power is a function of industry structure (few concentrated suppliers, differentiation, switching costs) and strategy – not wishful thinking by management.

$$\text{Revenue} = \text{Volume} \times \text{Price} \quad \Delta\text{Rev} \approx \% \Delta \text{Volume} + \% \Delta \text{Price} + \text{Mix} + \text{FX}$$

where: Mix = shift in product / geography weights; FX = currency translation for reporters in one functional currency.

CFA LM 5 SELF-ASSESSMENT Q4 – BEVERAGE PRICE × VOLUME WALK

Tea-based drink manufacturer: last year 2 m cases at USD 20 average price. This year: 3 m cases, ASP rises 5% (less discounting). Forecast net-sales growth?

1. Prior-year sales: $\text{USD } 20 \times 2 \text{ m} = \text{USD } 40 \text{ m}$
2. Forecast ASP: $\text{USD } 20 \times 1.05 = \text{USD } 21$
3. Forecast sales: $\text{USD } 21 \times 3 \text{ m} = \text{USD } 63 \text{ m}$
4. Growth: $63 / 40 - 1 = +57.5\%$, rounded to 58%

Growth = +58%. Volume effect +50%, price effect +5%, interaction +2.5% – all three at once.

Source: CFA Institute, Curriculum Vol 5, Learning Module 5, self-assessment Q4 (p. 199) and §4 "Revenue Analysis", pp. 217–223.

Operating and financial leverage multiply each other — total leverage is their product, and can make or break a cyclical business

Rule of thumb: a firm already running high operating leverage should be cautious with financial leverage — their product is what shareholders actually experience.

$$\text{DOL} = \frac{\% \Delta \text{EBIT}}{\% \Delta \text{Sales}}$$

$$\text{DFL} = \frac{\% \Delta \text{NI}}{\% \Delta \text{EBIT}}$$

$$\text{DTL} = \text{DOL} \times \text{DFL}$$

where: DOL = degree of operating leverage (from fixed costs). DFL = degree of financial leverage (from fixed interest). DTL = degree of total leverage.

CFA LM 5 SELF-ASSESSMENT Q6 — COMPANY XYZ DOL

Revenue 20X0: EUR 9,256 m → 20X1: EUR 9,707 m. Operating income: EUR 1,527 m → EUR 1,673 m. What is DOL for 20X1?

1. $\% \Delta \text{Sales} = 9,707 / 9,256 - 1 = +4.87\%$

2. $\% \Delta \text{EBIT} = 1,673 / 1,527 - 1 = +9.56\%$

3. $\text{DOL} = 9.56\% / 4.87\% = 1.96\times$

DOL = 1.96× — a 1% sales move becomes a 1.96% EBIT move. DFL on top amplifies it again at the equity line.

Source: CFA Institute, Curriculum Vol 5, Learning Module 5, self-assessment Q6 & Q8 (pp. 200–201); DOL §5 p. 226; DFL §6 p. 237; DTL p. 244.

SECTION 02

Industry & competitive analysis

Industry structural factors set a long-run profitability ceiling — company-specific factors create the spread beneath it

Competition "pulls" company profitability to an industry base rate over time. Industry analysis estimates that base rate; company analysis estimates the gap to it.

STEP 1

DEFINE

Industry boundaries, constituents, substitutes

STEP 2

SURVEY

Size, growth, character, profitability, market share

STEP 3

STRUCTURE

Porter Five Forces — long-run profitability drivers

STEP 4

EXTERNAL

PESTLE — growth & share dynamics

STEP 5

POSITION

Company strategy vs industry forces

WHY INDUSTRY MATTERS

Firms in the same industry face similar demand, factor costs, regulation, and substitutes. Structural forces (Porter) determine long-run returns on invested capital — the ceiling.

Company-specific factors — size, execution, strategy — determine how close a firm gets to that ceiling.

IMPROVED FORECASTS & IDEAS

Industry analysis sharpens forecasts by modelling the competitive forces that discipline prices, costs, and share.

It also surfaces investment candidates the bottom-up screen missed — the second-best in an attractive industry can beat the best in a structurally broken one.

¹ McGahan & Porter (1997): industry mattered most for the SUSTAINABILITY of economic profits — the forces act as a ceiling, not a floor.

Source: CFA Institute, Curriculum Vol 5, Learning Module 6, §2 "Uses of Industry Analysis", Exhibit 2, pp. 248–250. McGahan & Porter (1997).

GICS, ICB and TRBC are strict hierarchical taxonomies — useful starting points, but limited by multi-segment firms and geography

A "demand" grouping — similar products from the customer's perspective — has largely replaced legacy "supply" schemes like SIC / NACE / ISIC.

Scheme	Provider	Tiers (top → bottom)	Groups per tier
GICS	MSCI+S&P DJI	Sector → Ind. Group → Industry → Sub-Industry	11 / 25 / 74 / 163
ICB	FTSE Russell	Industry → Supersector → Sector → Subsector	11 / 20 / 45 / 173
TRBC	Refinitiv	Econ. Sector → Business → Ind. Group → ...	14 / 33 / 62 / 154
SIC / NACE / ISIC	Governments	Legacy supply-based – updated infrequently	varies

FOUR LIMITATIONS ANALYSTS LIVE WITH

1. Groups too narrow or too broad (Shopify sits with Check Point in "Application Software") · 2. Multi-product firms get a single bucket (Amazon is Consumer Discretionary although most operating profit is AWS) · 3. Global schemes miss local industries (healthcare services, where geography and licensing bind) · 4. Groupings change over time, breaking prior-period comparability (the Real Estate sector was carved out of Financials).

¹ Multi-segment firms: classify on the segment with >50% of revenue, profits or assets; failing that, ≥60% of revenue; failing that, discretion.

Source: CFA Institute, Curriculum Vol 5, Learning Module 6, §3 "Industry Classification", Exhibits 3–4, pp. 251–257.

Porter's Five Forces assesses long-run industry profitability — scored High / Moderate / Low on each of the five dimensions

Scoring is qualitative — if several forces are intense, almost no firm earns attractive returns on capital; if all are benign, the industry is a goldmine.

Force	Score	Driver — why this rating?
Threat of new entrants	VERY HIGH	>40,000 new US retail firm formations per month
Threat of substitutes	LOW	Services replace some spending, but not most goods
Bargaining power of customers	MODERATE	Fragmented buyers, but the internet made SKUs comparable
Bargaining power of suppliers	LOW-MODERATE	Many manufacturers; branded and luxury are the exception
Rivalry among competitors	HIGH	Many similar firms, similar SKUs — price is the weapon

VERDICT — THREE OF FIVE FORCES ARE HOSTILE

Very high entry threat, high rivalry, moderate customer power — structurally, retail caps returns on invested capital for the median operator. The escape routes are scale, scope and brand, and they run in opposite directions: Costco clears a low-single-digit operating margin on volume plus membership fees; a category specialist like Home Depot runs a double-digit one. Same forces, two answers.

Source: CFA Institute, Curriculum Vol 5, Learning Module 6, §5 Case Study "Porter's Five Forces for US Retail (ex autos)", pp. 270–271. Costco/Home Depot margins: presenter commentary.

An analyst must rank the five forces — one or two usually dominate, and those are what must be monitored quarter to quarter

"What would happen if company X raised prices by 10% tomorrow?" If a large share of customers would switch, buyer power and substitutes are the forces to watch.

BRANDED BEVERAGES

Dominant: New entrants — LOW

Incumbents hold shelf-space arrangements with retailers, own distribution, and lock up on-trade accounts. A new brand cannot reach the customer.

CLOUD INFRASTRUCTURE

Dominant: New entrants — LOW

Enormous fixed assets in servers and networking, near-zero variable cost, and economies of scope with adjacent businesses. Incumbents cut unit prices as they scale.

GENERIC PHARMA

Dominant: Rivalry — VERY HIGH

Regulators certify chemical equivalence, so pharmacies are indifferent. One generic competitor vs nine moves price by roughly 6× (Dave et al., 2017).

US COMMERCIAL BANKS

Dominant: New entrants — VERY LOW

Fewer than three new commercial bank charters per year over 2010–2021. Regulatory capital, deposit insurance and licensing make de novo entry near-impossible.

COMMERCIAL AIRCRAFT

Dominant: Rivalry — HIGH

A duopoly, yet every other force is benign and they still fight on price, financing and warranty terms per sale. Specialised assets make exit costly.

ENTERPRISE SOFTWARE

Dominant: Buyer power — LOW

SAP-class implementations take years and wire into mission-critical systems. Switching costs neutralise the buyer; incumbents price accordingly.

¹ Each label names one of the five forces; the phrase after it is the DRIVER that makes the force intense or benign there.

Source: CFA Institute, Curriculum Vol 5, Learning Module 6, §5 "Industry Structure", checklist and ranking guidance, pp. 265–270.

PESTLE catches what Porter misses — the external tides that move an industry's growth rate and share dynamics

Porter asks "how attractive is this industry?". PESTLE asks "what outside it is about to change?". Porter drives profitability; PESTLE drives growth and share.

Factor	What it covers	Beauty-industry example
P — Political	Policy, regulation, geopolitics	Excise tax debated on plastic packaging
E — Economic	GDP, income, inflation, rates, FX	EM income growth lifts premium demand
S — Social	Demographics, culture, lifestyles	Phone cameras and social media lift demand
T — Technological	Sustaining vs disruptive innovation	Influencer video: an efficient new channel
L — Legal	Court and policymaker rule changes	Counterfeit and trademark enforcement
E — Environmental	Carbon transition, waste, land use	Sustainable-packaging commitments

THE EXAM QUESTION: PESTLE OR PORTER?

A legislature debating a packaging tax is PESTLE — external, political. A rival launching AR "try-on" filters is NOT PESTLE: it is a competitor move inside the industry, so it is Porter. Test: is the actor inside the industry or outside it?

Source: CFA Institute, Curriculum Vol 5, Learning Module 6, §5 "External Influences on Industry Growth", pp. 271–274; beauty example from LM 6 self-assessment Q7, p. 248.

Three generic competitive strategies — cost leadership, differentiation, focus — explain most of what companies actually do

Evaluating a strategy has three tests: (1) does it defend against the five forces? (2) does it align with PESTLE? (3) does the firm have the resources and capabilities to execute?

COST LEADERSHIP

Lowest unit cost in the industry.

Means:

- Scale economies from fixed costs
- Favorable raw-material access
- Strict cost culture
- Aggressive pricing for volume
- Low-cost distribution
- Economies of scope

Exemplars:

Costco · Walmart · Ryanair · Southwest Airlines · Vanguard

DIFFERENTIATION

Uniqueness customers will pay for.

Means:

- Brand, advertising, proprietary channels
- Trademarks, patents, copyright
- Superior quality / unique features
- Culture of customer experience
- Integration of hardware, software, service
- Premium pricing

Exemplars:

Apple · LVMH · Rolex · Nike · Disney

FOCUS

One narrow segment, served exceptionally.

Means:

- Proximity to a specific customer group
- Deep understanding of their needs
- May borrow from cost leadership OR differentiation — but only inside that niche

Exemplars:

Ferrari (luxury auto) · Edward Jones (small-town US brokerage) · Ducati (sport motorcycles)

¹ "Stuck in the middle" — not a cost leader, not differentiated, not focused — is the worst place to be. Effectiveness is only ever judged in hindsight.

Source: CFA Institute, Curriculum Vol 5, Learning Module 6, §6 "Competitive Positioning", Exhibit 6, pp. 276–279. Porter (1980).

SECTION 03

Company forecasting

There are four forecast approaches — historical, base-rate convergence, management guidance, and analyst discretionary — each fits a different setting

In practice the four are combined — guidance for next year, base-rate convergence for the long run, discretionary judgement at a disruptive inflection.

Approach	Best-fit setting	Where it breaks
Historical results	Mature, non-cyclical; immaterial lines	Past conditions need not repeat
Base rates & convergence	Young firms maturing toward peer norms	Leader IS the base rate (Alphabet)
Management guidance	Good track record; firm-level variables	Macro lines — no info advantage
Analyst discretionary	Cyclicals, no comparables, disruption	Only as good as the judgement

AND FOUR FORECAST OBJECTS — PICK THE RIGHT GRANULARITY

- 1. DRIVERS of financial-statement lines (stores × sales per store) — highest explanatory value. · 2. INDIVIDUAL LINES directly — efficient where drivers are unclear or the line is immaterial. · 3. SUMMARY MEASURES (FCF, EPS, total assets) — efficient but opaque.
- 4. AD HOC objects (litigation, regulatory action, disaster) — not yet on any statement.

Rule: forecast only what is disclosed regularly — otherwise you can never check whether you were wrong.

¹ Ciconte et al. (2013): the UPPER bound of a guidance range often represents management's true expectation, padded with a beatable lower bound.

Source: CFA Institute, Curriculum Vol 5, Learning Module 7, §2 "Forecast Objects, Principles, and Approaches", Exhibit 1, pp. 288–294.

Top-down revenue forecasts pivot on market size and share; bottom-up forecasts build from volumes and prices — cross-checking both uncovers hidden assumptions

Run both on the same company. Where they disagree, the gap is an assumption you have not made explicit yet — usually market share.

TOP-DOWN (MARKET × SHARE)

Step 1 — forecast market size

- nominal GDP, then a premium or discount to it
- or market growth directly, adjusted for life cycle

Step 2 — forecast company share

- stable, gaining, losing — and by how much

Step 3 — multiply

- Revenue = Market size × Share

CFA example (Warehouse Club Inc.):

- US retail ex autos grew 3.4% p.a. over 10 years
- Share was 70 bps in 2X19, rising ~2 bps a year
- Forecast = market × 1.034 × (share + 2 bps)

BOTTOM-UP (VOLUME × PRICE)

Step 1 — forecast capacity / units

- stores opened, subscribers added, planes in fleet

Step 2 — forecast unit economics

- sales per store, ARPU, yield per seat-mile

Step 3 — multiply & aggregate

- Revenue = Σ (units × unit price) by segment

CFA example (Warehouse Club Inc.):

- 4–6 new stores opened per year
- Net sales PER STORE grew 3.8% p.a.
- ~450,000 members added per year × annual fee

Source: CFA Institute, Curriculum Vol 5, Learning Module 7, §3 "Forecasting Revenues", Exhibit 2 (Warehouse Club Inc.), pp. 296–298.

A three-scenario forecast — bear, base, bull — disciplines the analyst into quantifying the risks already identified in Porter and PESTLE

Every scenario needs a story, not just a number — bear = recession plus accelerating e-commerce share loss; bull = store traffic recovers and pricing sticks.

BASE CASE — ILLUSTRATIVE US CONSUMER-STAPLES RETAILER (FY + 1)

Prior-year revenue USD 11,500 m. Real GDP +2.0% and inflation +2.5% → nominal GDP +4.5%; industry tracks it. Bottom-up: same-store sales +3.5% and five new stores at USD 50 m each. Five stores open in every scenario.

1. Top-down: $11,500 \times 1.045 = 12,018$ m
2. Bottom-up: $11,500 \times 1.035 = 11,903$, $+ (5 \times 50) = 250 \rightarrow 12,153$ m
3. Gap = 135 m $\approx 1.1\%$ of sales — that gap IS an implied share gain. Split it: $(12,018 + 12,153) / 2$

BASE-CASE FY+1 REVENUE = USD 12,085 m (+5.1% YoY)

Scenario	Story	Nom. GDP	SSS	Top-down	Bottom-up	FY+1 rev
Bear (20%)	Recession; e-commerce takes share	+1.0%	−3.0%	11,615	11,405	11,510
Base (60%)	Trend holds; share edges up	+4.5%	+3.5%	12,018	12,153	12,085
Bull (20%)	Traffic recovers; pricing sticks	+6.0%	+6.0%	12,190	12,440	12,315
Weighted	Probability-weighted expectation					12,016

Source: CFA Institute, Curriculum Vol 5, Learning Module 7, §6 "Scenario Analysis", pp. 318–319. Figures here are an illustrative build, not a CFA example.

Scenario analysis of a disruptive threat — the iPad cannibalizing PC demand — sizes potential Microsoft revenue losses with explicit assumptions

Set at the start of 2012, just over a year after the iPad launched. The question: what share of PC sales do tablets replace, and what does that cost Microsoft?

Row (thousands of units unless noted)	FY11	FY12E	FY13E	FY14E	3-y CAGR
Pre-cannibalisation PC shipments	350,903	360,000	380,000	400,000	+4.5%
Tablet shipments	38,471	90,000	125,000	175,000	+65.7%
Cannib. factor, consumer / non-consumer	30% / 10%	30% / 10%	30% / 10%	30% / 10%	—
PCs cannibalised by tablets	11,205	25,560	34,750	47,250	—
MSFT revenue lost (USD m, ASP 85 / 155)	964	2,223	3,050	4,200	+63%

WHAT THE SCENARIO ACTUALLY BUYS YOU

Base case (30% / 10%): Microsoft loses ~USD 2.2 bn of revenue in FY12, rising to USD 4.2 bn by FY14 — and with ~70% of its cost base fixed, that FY14 hit becomes USD 2.6 bn of operating income, 96 bps of margin and USD 0.25 of EPS. Move the factors to 40% / 15% and FY12 goes from 2.2 to 3.0 bn. The call turns on which is priced in.

Source: CFA Institute, Curriculum Vol 5, Learning Module 7, Example 4, Exhibits 13 & 17 and the Knowledge Check, pp. 319–325.

SECTION 04

Valuation toolkit

Three families of valuation models — present-value, multiplier, and asset-based — together triangulate an intrinsic-value estimate

Price < value → undervalued; price > value → overvalued; price ≈ value → fairly valued — always interpreted with an analyst confidence band.

PRESENT VALUE (DCF)

Intrinsic value = PV of future cash flows to equity.

Models:

- Gordon growth DDM
- Multi-stage DDM
- FCFE model

Best for:

Mature, dividend-paying, predictable firms with forecastable cash flows.

Weakness:

Sensitive to r and g ; terminal value often dominates.

MULTIPLES

Intrinsic value inferred from comparable companies.

Equity level: P/E, P/B, P/S, P/CF

Enterprise level: EV/EBITDA, EV/Sales, EV/Operating income

Trailing or forward — either is fine, provided it is the SAME basis across every company compared.

Weakness:

Assumes peers are correctly priced; accounting noise; cyclicity distorts the denominator.

ASSET-BASED

Intrinsic value = Σ assets – Σ liabilities – preferred, at fair values.

Best for:

- Financial companies
- Natural-resource companies
- Former going concerns being liquidated

Weakness:

Ignores intangibles (brands, people, growth) and is hard where PP&E has no observable price. A floor, not a going-concern value.

The DDM prices a share as the PV of future dividends — FCFE applies the same idea to dividend-paying capacity instead of actual dividends

Mature dividend-payer with a stable payout? DDM. Young firm, non-payer, or heavy buybacks? FCFE. Both are equity-level, so both discount at the cost of equity.

$$V_0 = \sum D_t / (1 + r)^t$$

$$\text{FCFE} = \text{CFO} - \text{FCInv} + \text{Net borrowing}$$

where: D_t = expected dividend in year t ; r = required return on EQUITY; FCInv = capex; Net borrowing = new debt – repayments.

USE DDM WHEN

- The firm pays a stable, predictable dividend
- Dividends track earnings at a consistent payout
- Management has signaled dividend policy credibly
- Examples: mature utilities, consumer staples, tobacco

Pitfalls: very low or zero dividends force assumptions; buybacks distort the "dividend" proxy.

USE FCFE WHEN

- The firm doesn't pay dividends (Amazon; Meta pre-2024)
- Dividends are disconnected from capacity to pay
- Heavy buybacks supplement or replace dividends
- Leverage changes meaningfully affect equity CF

FCFE needs CFO, capex, working capital and net borrowing — more inputs, but far less arbitrary than guessing when a dividend starts.

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §5 "Dividend Discount Model (DDM) and Free-Cash-Flow-to-Equity Model (FCFE)", Eqs. 1 & 4, pp. 343–345.

The Gordon growth model prices a stock as a growing perpetuity — elegant, but extremely sensitive to r and g

Deck 01 did this on Shipline: $D_0 = 1.50$, $r = 15\%$, $g = 6\% \rightarrow \text{GBP } 17.67$. Same formula, different inputs — here CFA uses Siemens, $D_0 = 3.70$, $r = 7.5\%$, $g = 5.4\%$.

$$V_0 = D_0 \times (1 + g) / (r - g) = D_1 / (r - g)$$

where: Assumes $r > g$ (else infinite); constant g forever; constant r forever; dividends are the right cash-flow measure.

CFA LM 8 EXAMPLE 7 — SIEMENS AG (FY2017)

Most recent dividend $D_0 = \text{€}3.70$. Dividends grew 5.4% p.a. over 2013–2017 [$3.00(1+g)^4 = 3.70$]. Required return estimated at 7.5%.

1. $D_1 = 3.70 \times 1.054 = 3.90$
2. $r - g = 7.5\% - 5.4\% = 2.1\%$
3. $V_0 = 3.90 / 0.021$

$V_0 = \text{€}185.70$ vs market $\text{€}119.20 \rightarrow$ appears undervalued

$r \downarrow / g \rightarrow$	3.5%	4.5%	5.5%
7%	109	155	260
8%	85	110	156
9%	70	86	112
10%	59	70	87

¹ The numerator is D_1 , the NEXT-year dividend — either $D_0(1+g)$ or a forecast D_1 . Putting D_0 on top is the single most common error here.

² Sanity check g with $b \times \text{ROE}$: Siemens' 5-year average retention $\times$ ROE gives $\approx 0.49 \times 17.3\% \approx 8.5\%$, well above the 5.4% dividend-growth estimate.

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §7 "The Gordon Growth Model", Example 7 and Exhibit 4, pp. 350–352.

The two-stage DDM bolts a Gordon terminal onto an explicit high-growth phase — the workhorse for firms transitioning to maturity

IBM in 2018: dividends had grown 13.5% p.a. over five years, but Value Line forecast 4.5% for 2015–2023. Two-stage DDM is built for exactly that profile.

$$V_0 = \sum D_0(1+g_S)^t / (1+r)^t + [D_{(n+1)} / (r - g_L)] / (1+r)^n$$

where: g_S = stage-1 growth; g_L = long-run growth. $D_{\{n+1\}}$ on top puts the terminal value at end of year n — discount n periods, not $n+1$.

CFA LM 8 EXAMPLE 10 — IBM TWO-STAGE DDM (END 2018)

D_{2018} = USD 6.21. g_S = 5% for one year, then g_L = 2.33% forever (produces the 8-y CAGR of 4.5% Value Line forecasts). r from CAPM: $2\% + 0.95 \times 5\% = 6.75\%$. CFA carries two explicit dividends, so $n = 2$.

1. $D_{2019} = 6.21 \times 1.05 = 6.5205$ $D_{2020} = 6.5205 \times 1.0233 = 6.6724$
2. $D_{2021} = 6.6724 \times 1.0233 = 6.8279$ — the first dividend of the perpetual-growth phase
3. $V_{2020} = D_{2021} / (r - g_L) = 6.8279 / 0.0442 = 154.477$ — struck at the END of 2020
4. $V_{2018} = 6.5205/1.0675 + 6.6724/1.0675^2 + 154.477/1.0675^2 = \text{USD } 147.52$

$V_0 = \text{USD } 147.52$ vs market $\text{USD } 148.56 \rightarrow$ approximately fairly valued

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §8 "Multistage Dividend Discount Models", Example 10, pp. 355–357.

Free cash flow to equity is the capacity to pay dividends — start from net income, add non-cash, subtract net investment, add net borrowing

FCFE ≠ dividends paid. A firm generating USD 950 of FCFE and paying USD 300 of dividends is retaining USD 650 — for reinvestment, buybacks, or a future distribution.

$$\text{FCFE} = \text{NI} + \text{D\&A} - \Delta\text{WC} - \text{Capex} + \text{Net borrowing}$$

where: the same thing as CFO – FCInv + Net borrowing. FCFE is an EQUITY claim — discount at the cost of equity and the result IS equity value.

Build step	USD m	Why it belongs here
Net income	1,000	After-tax earnings, bottom of the P&L
+ Depreciation & amortisation	+400	Non-cash charge already deducted in NI
– Increase in working capital	–50	Growth ties up cash in receivables/stock
– Capital expenditure	–600	FCInv – needed to stay a going concern
+ Net borrowing	+200	New debt is cash available to equity
FCFE available to common shareholders	950	95% of net income – healthy conversion

¹ $V_0 = \sum \text{FCFE}_t / (1+r)^t$ with a Gordon tail. Discount FCFE at WACC, or subtract net debt from an FCFE value, and you double-count the debt.

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §5 "FCFE Model", Eq. 4, pp. 344–345. Figures here are an illustrative build.

The justified forward P/E falls out of the Gordon model as payout / (r – g) — which is why high-quality, high-growth, low-r firms print high P/Es

Watch the trap: raise the payout and the numerator rises, but retention falls, so $g = b \times \text{ROE}$ falls and the denominator widens. Never move one input alone.

$$P_0/E_1 = \text{payout} / (r - g) \qquad g = (1 - \text{payout}) \times \text{ROE}$$

CFA EX. 12 — NESTLÉ JUSTIFIED FWD P/E

Payout ratio 68% (2013–17 average, rounded); retention $b = 32\%$; long-run ROE estimated at 21.5%; required return $r = 9\%$.

1. $g = b \times \text{ROE} = 0.32 \times 0.215 = 6.9\%$
2. $r - g = 9.0\% - 6.9\% = 2.1\%$
3. $P_0/E_1 = 0.68 / 0.021$
4. Sell-side forward P/Es at the time: 19.3× to 26.3×

Justified forward P/E = 32.4× — above the whole street

Payout ↓ / g →	5.0%	6.0%	7.0%
55%	13.8	18.3	27.5
65%	16.3	21.7	32.5
70%	17.5	23.3	35.0
75%	18.8	25.0	37.5

¹ FORWARD P/E uses next year's EPS, TRAILING the last twelve months. Either works — but every company in a comp set must use the same basis.

² Grid assumes $r = 9\%$. Raising r to 9.5% cuts the justified P/E to 26.2×; using ROE of 16.5% instead of 21.5% cuts it to 18.4×.

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §9 "Multiplier Models...and Fundamentals", Example 12 and Exhibit 6, pp. 360–362.

Method-of-comparables P/S ranks autos in one line — but cheap multiples demand a "why" before a "buy"

Law of one price: identical assets should sell for the same price. Cars are not identical — growth, margin, leverage and cyclicity all differ. P/S is where the question starts.

Automaker	P/S	What you would have to check before calling it cheap
Peugeot	0.28	Lowest P/S — mid-turnaround, thin and volatile margins
Ford Motor	0.33	US cyclical exposure; legacy obligations sit ahead of equity
General Motors	0.36	Like Ford, on a balance sheet cleaned up after 2009
Nissan Motor	0.38	Alliance structure and governance complicate the equity story
Honda Motor	0.46	Diversified beyond cars — motorcycles and power products
Tata Motors	0.49	India volume plus Jaguar Land Rover — two different businesses
Daimler	0.55	Premium mix (Mercedes-Benz) supports a higher multiple
BMW	0.57	Premium-only mix, strong brand, high retention
Toyota Motor	0.80	Highest P/S — scale, reliability, capital discipline

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §10 "Method of Comparables", Example 13, p. 363 (P/S as at December 2017). Right-hand column is presenter commentary, not CFA text.

EV/EBITDA compares firms at the operating level, immune to capital-structure noise — the metric of choice for cross-border and LBO work

EV is the whole capital stack an acquirer pays for — equity + preferred + debt – cash. EBITDA is pre-D&A, pre-interest, pre-tax: the cash engine, before financing and accounting choices.

$$\text{EV} = \text{Market cap} + \text{Preferred} + \text{Debt} - \text{Cash}$$

where: Debt at MARKET value where available — book value is only a rough proxy, since rates and perceived credit risk move after issue. Cash includes short-term investments.

CFA EX. 16 — CAMECO (FY2017, CAD m)

395.8 m shares at CAD 14.11. LT debt: book 1,905, market 1,480.
Other liabilities 1,014. Cash 592. EBITDA 606.

1. Market cap = $395.8 \times 14.11 = 5,585$
2. Debt & other liabilities = $1,480 + 1,014 = 2,494$
3. EV = $5,585 + 2,494 - 592 = 7,487$
4. EV / EBITDA = $7,487 / 606$

$$\text{EV} / \text{EBITDA} = 12.4\times$$

THE CONSISTENCY RULE

Numerator and denominator must describe the SAME claim.

EQUITY: price over EPS, book value, sales.

ENTERPRISE: EV over EBITDA, EBIT, revenue.

EV/EPS and price/EBITDA are the classic error. To cross over, subtract debt and preferred from EV.

¹ EV is the cost of a takeover: the acquirer assumes the debt and receives the cash. That is why cash is subtracted.

² EV survives what P/E cannot — different leverage, different tax regimes, negative earnings. But EBITDA adds back D&A, so it flatters capital-heavy firms.

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §11 "Enterprise Value", Example 16 and Exhibit 9, pp. 368–369.

Asset-based valuation sets a floor — useful for asset-heavy, intangible-light, or winding-down businesses, but weak when value sits in brands, people, or growth

Value = fair value of assets – fair value of liabilities – preferred. Getting from book to fair value is the whole difficulty, and excluded intangibles make the answer a floor.

MOST TRACTABLE — FAMILY LAUNDRY

Tangibles dominate: building, land, washers and dryers, inventory. Take two market appraisals on the property and cost the used machines at replacement.

Deduct the USD 100,000 loan.

But CFA is blunt: any intangible value — prime location, clever marketing — is excluded, which makes even this valuation inaccurate.

FLOOR ONLY — SECRET-RECIPE RESTAURANT

Equipment plus inventory, less liabilities, gives a baseline — and the space is rented, so there is no property to appraise.

The secret recipes and the owner's cooking are the business, and the owner is retiring. Some or all of the intangibles vanish on sale.

A minimum valuation, nothing more.

RIGHT LENS — LOSS-MAKING AIRLINE

Few routes, high labour costs, no dividend, losing millions a year. Most valuation approaches return a negative number.

But the routes, landing rights, airport facility leases, ground equipment and aircraft are worth a great deal to a competitor.

Value the assets apart from the business losing money with them.

¹ Hardest where it is most needed: firms with heavy PP&E that has no observable market price, and any hyper-inflationary environment.

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §12 "Asset-Based Valuation", Examples 18–20, pp. 371–372.

Deck 12 — Equity II — Analysis & Valuation

- An initial research report has eight elements, built on three analytical blocks — company, industry, forecast — with valuation and a recommendation on top.

- Porter sets the long-run profitability ceiling; PESTLE drives growth and share. Rank the forces — one or two always dominate, and those are what you monitor.

- Bear / base / bull prices the risks Porter and PESTLE found. CFA's iPad case: USD 2.2 bn of FY12 Microsoft revenue, and USD 0.25 of EPS by FY14.

- FCFE is dividend-paying capacity, discounted at the cost of equity for equity value. FCFF goes at WACC to enterprise value. Never cross the two rates.

- Business model = five questions: what is sold, to whom, through which channel, at what price, bought from whom. Then split revenue into volume $\times$ price $\times$ mix $\times$ FX.

- Four forecast approaches, four forecast objects. Triangulate top-down (market $\times$ share) against bottom-up (units $\times$ price): the gap is an implied share assumption.

- Gordon uses D_1 , not D_0 — Siemens €185.70. Two-stage bolts a Gordon tail on: IBM USD 147.52, 92% of it terminal, discounted n periods and never $n+1$.

- Justified forward P/E = payout/($r - g$): Nestlé 32.4 $\times$. EV/EBITDA (Cameco 12.4 $\times$) survives leverage. Keep numerator and denominator on the same claim.